

Flight Control System Modeling with SysML to Support Validation, Qualification and Certification

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Abstract: The introduction of new technologies to build ‘More Electric Aircraft’ induces new challenges for both the design and safety analysis of new aircraft. A model-based approach is needed for both design and validation processes in order to manage the complexity and validate the conformance to safety requirements. In this paper, a SysML-based approach merging MBSE and MBSA is presented. This approach is applied to a Flight Control System (FCS) both for the design and the validation processes. A parallel is made to compare the models used in each of these processes. This comparison led to suggestions of improvements both for the design and verification and validation approaches.

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1. INTRODUCTION

Most of the actuation systems in the aircraft are supplied by hydraulic power characterized with a poor global efficiency, and frequent maintenance operations. To cope with the drawbacks of the use of hydraulic technology, research works are focused on ‘More Electric Aircraft’ (see Derrien (2012), Reyssset (2015)). In this scope, the Flight Control Systems (FCS) are progressively relying on electric energy to replace all or part of the hydraulic systems for the actuation of the flight control surfaces. The introduction of such new FCS implies the use of new technologies with new actuators and on-board control unit. This will result in new failure modes that are not mastered because of the lack of feedback from experience. New challenges are then faced in the design as well as in the verification, validation and qualification of these systems in compliance with aeronautics safety standards such as the ARP 4761 SAE-Aerospace (1996).

Model-Based Systems Engineering (MBSE) is becoming compulsory for the design of such complex systems to help in better understanding and mastering these new technologies. It also helps in making the design more efficient, easier and faster and as a consequence, reducing design time and cost. Integrating safety aspects as early as possible in the design process helps in reducing risks (see Berres et al. (2015)). The early integration of safety in the design process is made possible by extending the system models with safety aspects facilitating the generation of safety analysis artifacts and reducing the gap between the design and safety analyses. Such extended model is useful both for the design and for the verification, validation and

qualification activities. Indeed, certification is currently based on huge set of documents provided by the aircraft manufacturers. Separate documents are used to describe the system and to detail the results of safety analyses (i.e. Functional Hazard Assessment and System Safety Assessment).

During the design stage, safety analyses aim at providing safety requirements that shall be met by the design team. As systems are getting more complex, it becomes almost impossible for a single safety expert to have a deep understanding of the whole system and all the technologies involved. Instead, a multi-disciplinary team is needed for safety and reliability analyses and this team shall collaborate with the designers to meet the safety requirements. A system model is then needed around which the multi-disciplinary team shall collaborate. This model should be built in a general system language understandable by all the team.

At the validation and qualification stage, a system model is also built, with different tools in order to validate that the safety requirements are well met and that the designed system complies with safety standards. In this paper, an integrated design approach including safety analysis using SysML language is presented. This approach is tested in a validation and qualification process. The different models used in each process are compared. A flight control system is used as an example to illustrate this work.

This paper is organized as follows. First, a state of the art about the integration of the MBSE-MBSA is given in section 2. Then, the FCS used as a case study example is described in section 3. Both the validation approach

applied in the DGA-TA and SafeSysE are respectively presented in section 4 and section 5. A discussion with improvements suggestion for both approaches is given in section 7. The paper is finally concluded in section 8.

2. RELATED WORK

Model Based Safety Analysis (MBSA) aims to provide a model-based approach to perform safety analyses while seeking a tighter integration between safety artifacts and design models. In this approach, system and safety engineers share a common system model created using a model-based development process. Joshi et al. (2006) proposed to augment the nominal system behavior captured in Simulink model-based development with the fault behavior of the system. To illustrate the process, they studied the Wheel Brake System as described in ARP 4761 Appendix L (SAE-Aerospace, 1996). The fault model consists of different component failures, i.e. digital and mechanical failure modes. Fault tolerance verification is carried out by using additional variables and real-time temporal logic operators to investigate if the system can handle some fixed number of faults. Informal safety requirements are formalized by temporal logic, and the model checker NuSMV is used to validate these requirements. Nevertheless, research challenges must be addressed on the choice of languages and tools, as well as the scalability of the analysis tool to cope with realistic systems.

AltaRica (Point, 2000) is an event-based modeling language which is designed to specify the behavior of complex systems. Mathematically based on Guarded Transition Systems, an AltaRica model is composed of nodes that are characterized by their reachable states, *in* and *out* flows, events, transitions and assertions. Once a system model is specified in the AltaRica language, it can be compiled into a lower level formalism such as finite-state machines, fault trees, stochastic Petri Nets or Markov chains (Cherfi et al., 2014; Mortada et al., 2104). The language is widely used for safety assessment of automotive, avionic and transport applications. It is supported by industrial tools such as Simfia, Cecilia OCAS and open source tools like OpenAltaRica (OpenAltaRica, 2015) with a graphical interface to design models, to inject failures and to simulate models. In Morel (2014), a model-based safety approach for early validation of avionics architectures is proposed. The model building contains four different levels: the Functional Hazard Analysis (FHA) view, the functional view, the physical view and the allocation. However, there are still some issues concerning the validation and the completeness of the allocation of functions in the functional view to hardware modules in the physical view. Model-based systems engineering with SysML can facilitate this task by building an allocation matrix since the early design phases.

Algorithms and translation rules allowing transformation from SysML diagrams to AltaRica Data Flow language have been proposed in different research works (Cressent et al., 2011; Ruin et al., 2012; Yakymets et al., 2013). Cressent et al. (2011) proposed a mapping between SysML models and AltaRica Data Flow (ADF) language, based on the MéDISIS framework. The first step is the translation of the SysML model to obtain the ADF description of the functional view of the system. The second step is the

modeling of the dysfunctional view using the Dysfunctional Behavior Database built and updated via different safety analyses such as Failure Mode and Effects Analysis (FMEA). Via AltaRica, existing tools to quantify reliability indicators such as the global failure rate, the mean time to failure, etc. can be used directly on the failure modes identified in different steps of MéDISIS. However, a complete automation of the translation between SysML and ADF language is not possible if some strict SysML construction rules such as expressive allocations between the modeling elements are not applied. As mentioned by the authors, some divergent declaration philosophies between the two languages (although sharing the object-oriented paradigm) impose complicated translation rules.

3. CASE STUDY DESCRIPTION

In this paper, we will study the flight control system (FCS) in civil aircraft. The flight controls of an aircraft typically include primary controls that govern the pitch, yaw and roll attitudes and the trajectory of the airplane as well as secondary controls dedicated to control the lift of the wings. The flight control surfaces of the civil aircraft Airbus A380 are illustrated in Fig. 1.

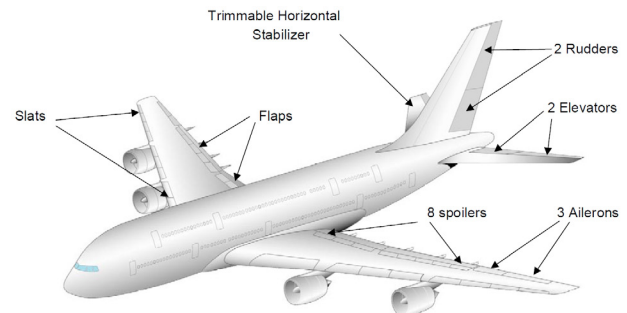


Fig. 1. A380 Flight Control System Architecture (Van den Bossche (2006))

The architecture of the flight control system is given by Fig. 2. The number of actuators per surface, as well as the number and distribution of power sources and flight control computers, are mainly imposed by safety considerations.

The FCS is made up of the actuators, sensors and on-board calculators. The flight control surfaces are actuated with different kinds of actuators powered by two different hydraulic circuits (Green circuit G and Yellow circuit Y) and two different electric circuits (E1 and E2). Each surface is actuated with one or two actuators powered by separate power sources. For instance, each side of the aircraft contains three ailerons O/B, M and I/B. The outer aileron (O/B) is actuated with two servocontrols powered by the Green and Yellow circuits respectively. The middle aileron (M) is actuated with a servocontrol and an Electro-Hydrostatic Actuator (EHA) powered respectively with the Green hydraulic circuit and the E2 electric circuit. The inner aileron (I/B) is actuated by a servocontrol and an EHA powered respectively with the Yellow circuit and the E1 electric circuit.

The flight control surfaces are considered outside the system. The power sources i.e. hydraulic and electric power

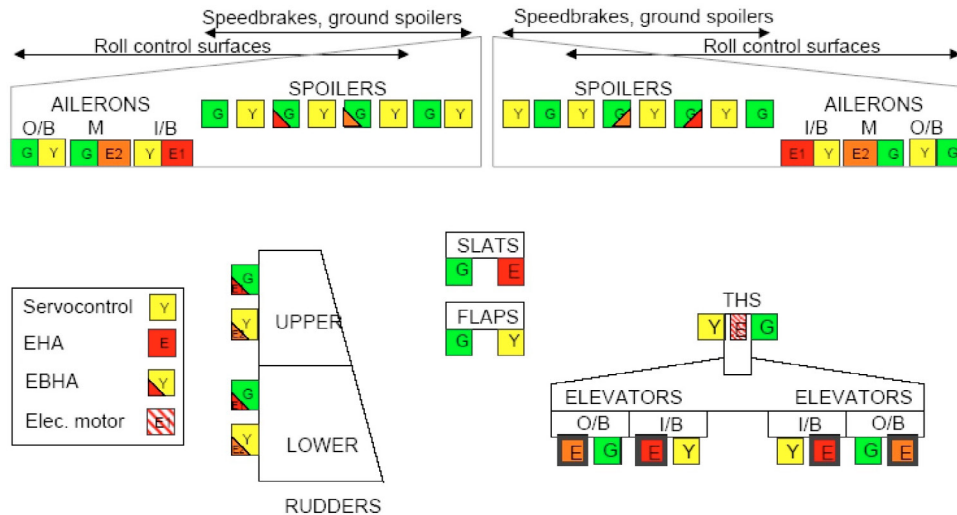


Fig. 2. Flight Control Surfaces of the A380 Aircraft (Van den Bossche (2006))

sources are also not part of the system since they are shared with other systems.

4. CERTIFICATION AND QUALIFICATION PROCESSES

This work was performed together with the Direction Générale de l'Armement - Techniques Avancées (DGA-TA) which is a french expertise center for ground tests of aeronautics systems and equipment. Among its missions, the DGA-TA is charged of the certification of civil aircraft in collaboration with the Direction Générale de l'Aviation Civile (DGAC) and the European Aviation Safety Agency (EASA). The SIE division of the DGA-TA is charged with the qualification and certification of critical systems and embedded software based on its expertise in the domain.

For its validation, qualification and certification activities, the DGA-TA uses model-based approaches with formal tools/languages such as AltaRica to assess the dysfunctional aspects. The AltaRica models represent the system in four different but consistent views:

- **Undesired events view:** this view contains the undesired events. An undesired event corresponds to the loss of one or several functions (i.e. one or several function that are no longer performed). The undesired events view contains state observers for some functions of the functional view.
- **Functional view:** this view contains the system functions achieved by the system and their breakdown into sub-functions. These functions are presented in a hierarchical way including logical operators to link a function at a certain level to the functions of the level below. Through these logical operators, the functional view also contains the failures propagation logic. Each function can have one of three states: 'Nominal', 'Degraded' or 'Loss'. The low-level functions are allocated to components from the component view.
- **Structural view:** the structural view models the interactions and the propagation logic of the failures among structural elements. Two levels of abstraction of the structural elements can be distinguished:

equipment level and component level. An equipment is made up of a set of components. An equipment has the same intrinsic states as a function and can be either in 'Nominal', 'Degraded' or 'Loss' state. A component can be either in a 'Nominal' state, a 'Loss' state (if it does not emit any signal or if it is detected to be erroneous) or a 'Misleading' state (erroneous but undetected state). Components failures can be simulated to assess their propagation and their effect at the system level. Each component is assigned to one of the airplane zones represented in the zonal view.

- **Zonal view:** the zonal view delimits the different airplane zones. Partitioning the airplane into different zones allows an additional safety measure by distributing the critical redundant components in different locations since some external threats such as electromagnetic radiation, or other internal accidents such as engine explosion can damage an entire zone of the aircraft. The zonal view helps in distributing the critical components in such a way that, the damage of one zone does not prevent to the fulfillment of any critical function.

Once the model for the system is built, a series of simulations is performed to assess the safety requirements. Component failures are progressively selected from the list of failures of each component, and the simulation propagates these failures to show their effects and if they induce any of the undesired events in the model. Fault trees can be generated from this model and minimal cut-sets and reliability studies can then be performed with appropriate tools.

This model-based approach allows a rigorous testing of the system behavior in presence of faults and allows a thorough validation of the safety requirements needed for the certification and qualification of the system/equipment in question. However, building the model is very long since it is based on descriptive documents provided by the airplane manufacturer. The provided documentation does not facilitate the understanding of the system. Indeed, textual descriptions are error prone and may have different

interpretations. They also may have inconsistencies as the same system may be described at more than one sections and the updates do not necessarily cover all the citations. Thus, the DGA-TA wishes to evaluate an integrated MBSE/MBSA methodology. A system model with a systems engineering dedicated language will help in having a better understanding of the system, prior to the safety assessment.

5. OVERVIEW OF THE INTEGRATED MBSE/MBSA PROCESS

This section gives an overview of the integrated process of systems engineering and safety analysis called SafeSysE. This integrated process is intended for the integration of safety analysis since the early design stages. For this, it extends a SysML-based systems engineering methodology presented in Mhenni et al. (2014) with safety analysis processes. The early integration of safety aspects influences the decisions taken during the design and avoids late design changes that are very costly and time consuming.

SafeSysE is described in Fig. 3 detailing the sequencing of the different processes of this approach (represented by SysML activities) as well as the exchanges between them. Data stores are used to model the storage of the different artifacts issued from each activity. Swim-lanes (columns) are used to make a distinction between systems engineering and safety analysis processes. For more detail about SafeSysE, please refer to Mhenni (2014).

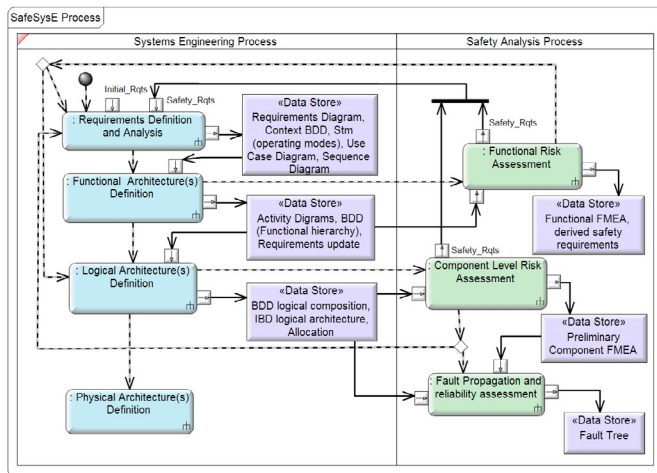


Fig. 3. SafeSysE Integrated Process

6. FLIGHT CONTROL SYSTEM MODELING USING SYSLML

In this section, the SafeSysE approach will be applied for the FCS. The focus will be mainly on the systems engineering part of the model in order to see how this can be helpful for the validation and certification processes. This study consists in reverse engineering since it is applied to an already existing system.

In a top-down design process, the system design begins with an external view, also called black box view of the system. In this phase, the system is considered as a black-box and the focus is on its external behavior and interactions. The aim of this part is to thoroughly define the

system specifications before moving to the second phase, where the white-box view is considered during the progressive definition of the solution. Each of the external and internal views contains structure, behavior and requirements representations. The resulting system model with all these views and aspects provide means to understand both the behavior and structure of the system and to link them together with their respective requirements. Having such a model instead of or in addition to the description documents provided for the certification team will be of a great help for the good understanding of the system and will reduce the inconsistencies and ambiguity of the natural language text document. However, as the intent of this methodology is mainly the design of the system, some views may contain irrelevant data for the validation and certification phase.

In the following, the system model for the FCS system will be given. The focus will be only on the useful views for the validation/certification phase. The first step in modeling a system is to define the system boundary. For this purpose, a Block Definition Diagram (BDD) in SysML is used to model the system context that describes the system as a black-box and its interactions with the different external elements. The context diagram of the FCS is given in Fig. 4 and shows the different external elements (represented as actors in SysML with the stick-man symbol) that interact with the system and their roles with regard to the system and/or the role of the system with regard to external elements. As an example, this diagram shows that the 'Control Surfaces' are controlled by the system (first role) and provide their position to the system (second role). It also shows that there are several types of 'Control Surfaces' such as 'Left Spoiler 1', 'Rudder', 'THS' etc.

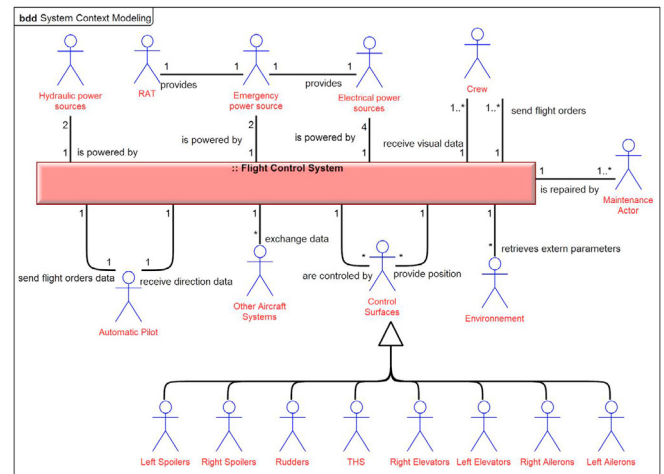


Fig. 4. Flight Control System Context

SysML offers a means to model the functional breakdown of the system. The first level functions of the FCS and their interactions are given in the activity diagram in Fig. 5. This diagram shows how the different external input flows (received by the external actors represented in the context diagram) are progressively transformed to provide the required output flows. Each sub-function can also be detailed in an activity diagram containing its sub-functions. A hierarchical representation can also be generated from this breakdown in a BDD as the one

in Fig. 6. This representation can be done for as many levels as the model contains. However, this diagram is simplified by giving only the first-level breakdown for a better readability.

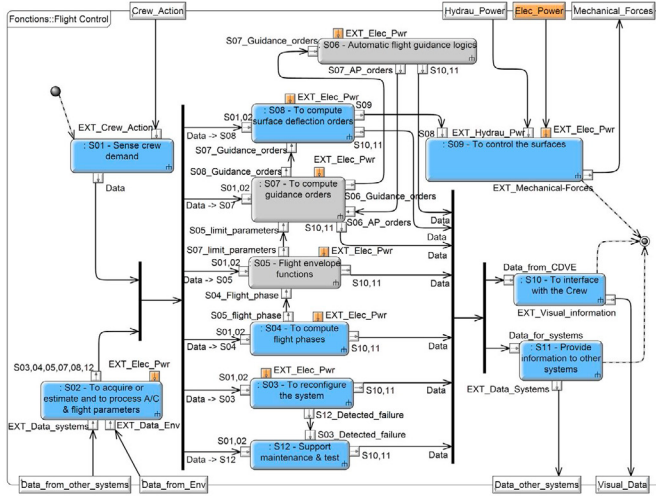


Fig. 5. Activity Diagram - Flight Control System Functions

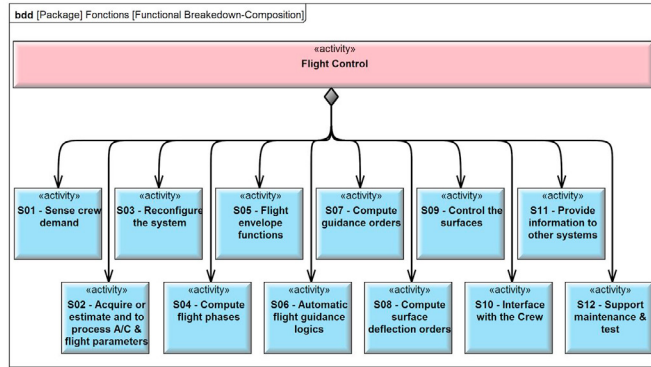


Fig. 6. Flight Control System: First Level Functional Breakdown

The breakdown of the 'S09 - Control the surfaces' function is given in Fig. 7.

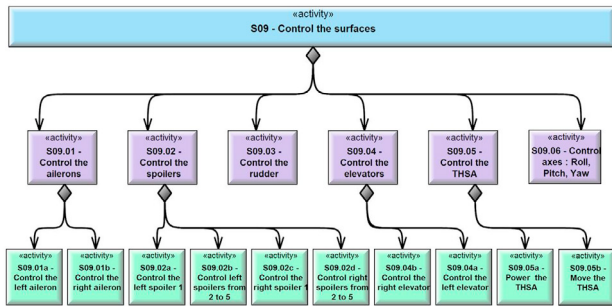


Fig. 7. Function - S09 Breakdown

After the functional architecture is defined, components are chosen and allocated to the identified functions. A BDD is then used to show the system composition (i.e. the different components of which the system is made up). The allocations can be represented either in an allocation matrix or in the mentioned BDD. An extract

of the composition BDD, focused on the components that actuate the Trimmable Horizontal Stabilizer (THS) is given in Fig. 8. The functions allocated to each component are mentioned in a compartment of the component block.

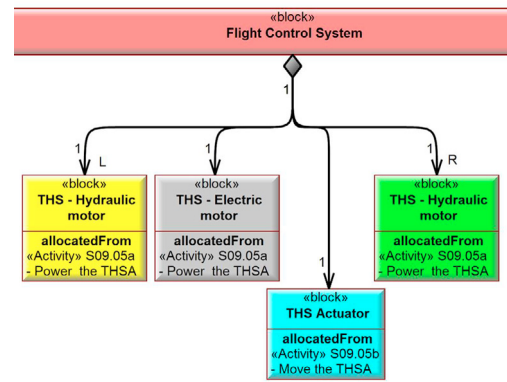


Fig. 8. Extract of the System Components and Functional Allocation

The interactions among the components are then modeled in an Internal Block Diagram (IBD). The interactions among components help in determining the error propagation through the components. However, unlike the components view in the AltaRica model, it does not include any logic operator for error propagation modeling. An extract of the IBD for the FCS is given in Fig. 9.

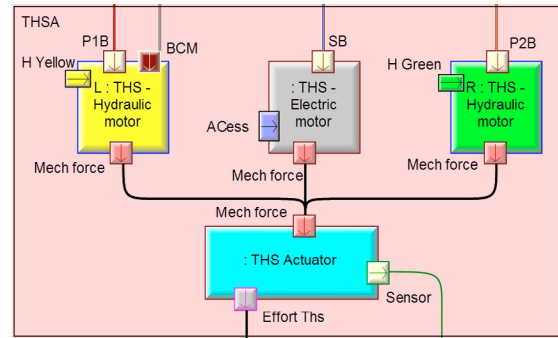


Fig. 9. Extract of the IBD for the THS related Components

7. DISCUSSION

This paper presents the results of an internship aiming at testing SafeSysE on a real industrial example in the DGA-TA and comparing it with the current methods used for safety analysis. The focus was given on the different models on which the safety analyses will be based. In SafeSysE, safety artifacts are generated from SysML models while in the DGA-TA AltaRica models are built for this purpose. As a result of this work, some improvements to both approaches are identified.

- Improvements for SafeSysE: As noticed through the different views presented earlier, the SysML-based modeling in SafeSysE does not provide any zonal view or any information about the location of the components within the whole system. The introduction of such information is very important for safety analysis and shall be considered for the improvement of SafeSysE. In SafeSysE approach, the undesired

events are not clearly classified in the model. Only the undesirable system level effects caused by functions or components failures, and identified in the FMEA process are stored as an attribute linked to the failure modes of each function or component respectively. This point shall also be considered. The third important point is the integration of logical operators to describe the functional as well as dysfunctional logic of exchanges between the system functions or components. This will help to introduce more dynamic aspects in the fault tree generation based on the functional system structure.

- Improvements for the DGA approach: On the other hand, some of the system views given by SafeSysE are missing in the current approach applied in the DGA-TA and, if provided, would help in a deeper and faster understanding of the system. As mentioned earlier, the documents provided by the aircraft manufacturers do not provide any global view of the system components nor of the system functions. The hierarchical views at the functional and structural levels given in SafeSysE are very useful to make sure of having the exhaustive list of functions and components with a clear allocation relationships between both. This will also avoid the inconsistencies and incompleteness in the names or descriptions of the functions and components that can be found in document-based description.

8. CONCLUSION

This study aims at evaluating the use of SysML language in the validation, qualification and certification process for aircraft systems, such as the flight control system. At DGA-TA, the implementation of the under development SysML-based MBSE-MBSA methodology called SafeSysE, has shown that some SysML structural diagrams (context, functions and components structures) may improve the whole understanding of the studied system as well as the consistency of its modeling, prior to the system safety-oriented analysis usually performed with AltaRica language. On the other hand, the simulation capability as well as some views in the AltaRica model could be integrated in SafeSysE to enhance the safety analysis part of this approach.

In addition, this study confirms the usefulness of sharing system models, using a common general purpose language such as SysML, between the aircraft manufacturers and their providers all along the design process. Indeed, this may help performing a thorough and consistent qualification and certification when they are made available to certifying entities such as DGA-TA.

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